TELECOM SALES ANALYTICS REPORT

By Timothy Mbilinya

(Business Intelligence Analyst)

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1. Introduction

This report presents an analysis of telecom sales performance using simulated data inspired by Airtel Malawi operations from 2022 to 2025. The purpose of the analysis is to evaluate sales trends, measure key performance indicators (KPIs), and generate insights that can support decision-making in sales strategy, revenue growth, and channel optimization.

The analysis was conducted using Python for data preparation, Excel for pivot-based analysis, and Power BI for interactive dashboard development.

2. Objectives

The main objectives of this analysis are:

- To analyze sales performance across regions, channels, and products
- To identify key trends in revenue and sales volume over time
- To evaluate the efficiency of different sales channels
- To provide actionable insights to improve business performance

3. Data Overview

The dataset contains telecom sales transactions from 2022 to 2025 and includes the following key variables:

- Date (transaction date)
- Region (North, Central, South)
- Channel (Agent, Distributor, Shop)
- Product (Airtime, Data Bundle, SIM)
- Units Sold
- Revenue (MWK)
- Commission (MWK)

The data was cleaned and processed using Python. Additional fields such as year and month were created to support time-based analysis.

4. Methodology

The analysis followed three main steps:

4.1 Data Cleaning

- Converted date to proper format
- Removed duplicates and checked for missing values
- Created new features such as Year and Month

4.2 Data Analysis

- Aggregated data by region, channel, and product
- Calculated key KPIs such as total revenue and commission rate
- Used pivot tables in Excel to summarize performance

4.3 Data Visualization

- Created charts in Excel (pivot charts)
- Developed an interactive dashboard in Power BI

5. Key Performance Indicators (KPIs)

The following KPIs were used:

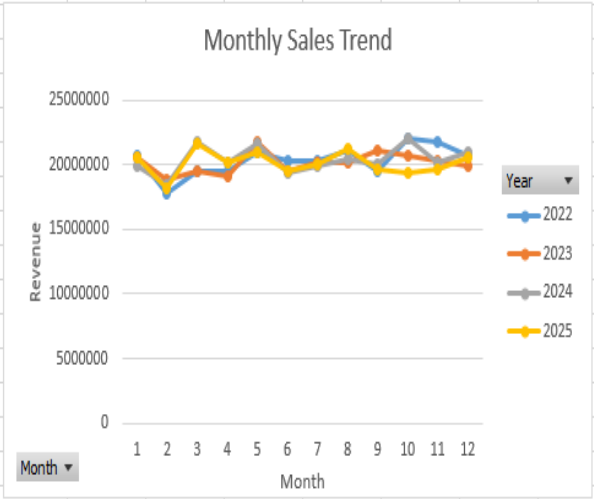
- Total Revenue
- Total Units Sold
- Commission Rate
- Revenue by Region
- Revenue by Channel
- Revenue by Product

These KPIs help measure both performance and efficiency.

6. Analysis and Findings

6.1 Monthly Sales Trend

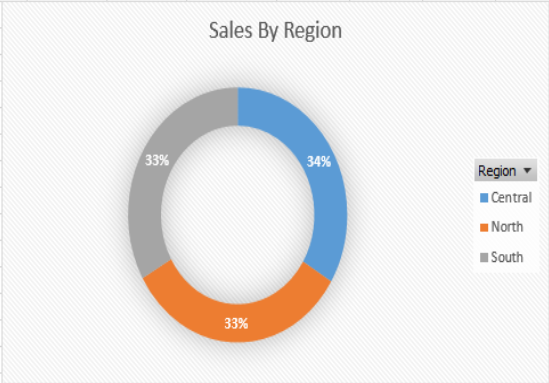
Sum of Revenu	Column Labels				
Row Labels	2022	2023	2024	2025	Grand Total
1	20704200	20618000	19864900	20522500	81709600
2	17845500	18910300	18502900	18186100	73444800
3	19582200	19522400	21806500	21620600	82531700
4	19457500	19112700	20248200	20142600	78961000
5	20922700	21810300	21664900	21025100	85423000
6	20274100	19584300	19374800	19471900	78705100
7	20368700	20226400	19953000	20101000	80649100
8	21087200	20173600	20450000	21217400	82928200
9	19518600	21090900	20038700	19701700	80349900
10	22111800	20718000	22109600	19357700	84297100
11	21731100	20338100	20168800	19674900	81912900
12	20695800	19980000	21046900	20598900	82321600
Grand Total	244299400	242085000	245229200	241620400	973234000



The monthly trend shows a steady increase in revenue over time, indicating business growth. Some months show higher sales than others, suggesting seasonal patterns in customer demand.

6.2 Regional Performance

Row Labels	Sum of Revenue_MWK
Central	328364700
North	321280500
South	323588800
Grand Total	973234000



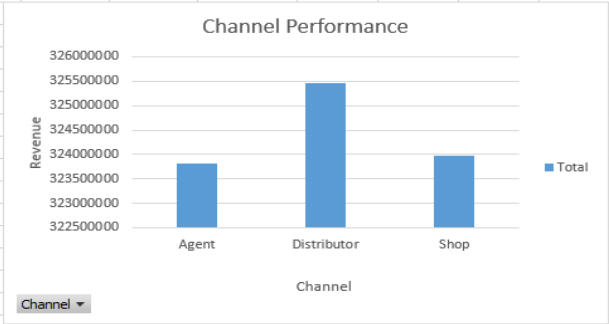
The Southern region generates the highest revenue, followed by the Central region, while the Northern region contributes the least.

This may be due to:

- Higher population and economic activity in the South
- Better distribution networks

6.3 Channel Performance

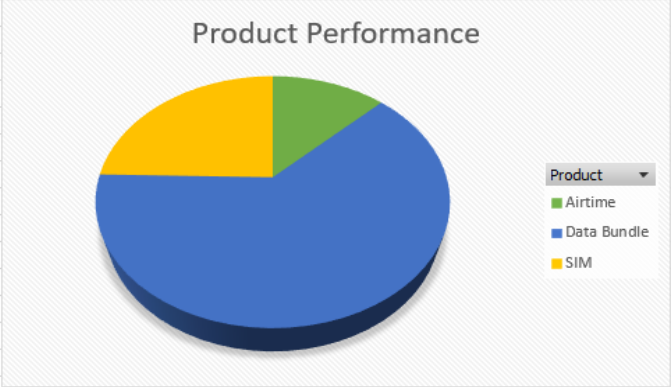
Row Labels	Sum of Revenue_MWK
Agent	323805400
Distributor	325461200
Shop	323967400
Grand Total	**973234000**



Distributors generate the highest revenue compared to Agents and Shops. This suggests that distributors play a key role in driving sales, likely due to wider reach and bulk transactions. Shops show lower performance, indicating a potential area for improvement.

6.4 Product Performance

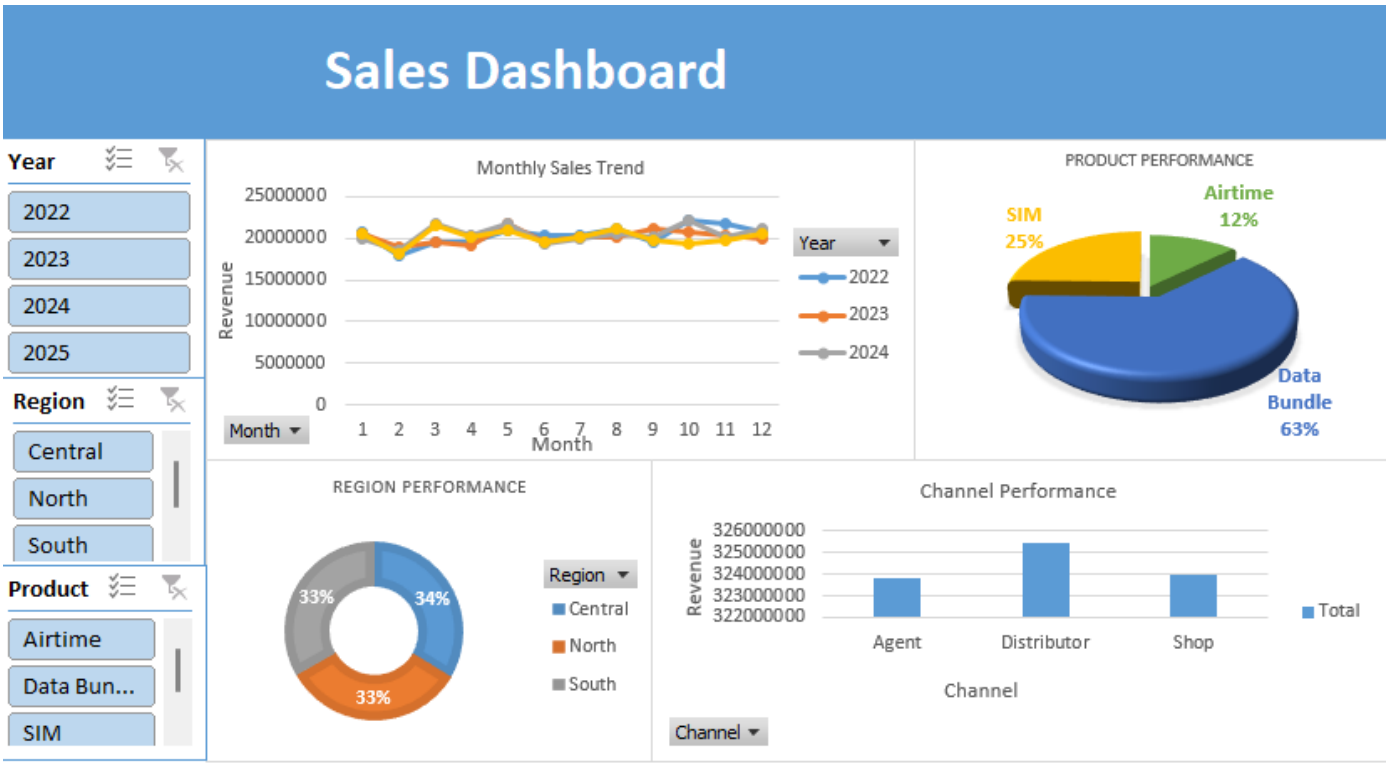
Row Labels	Sum of Revenue_MWK
Airtime	122648900
Data Bundle	610894500
SIM	239690600
Grand Total	**973234000**



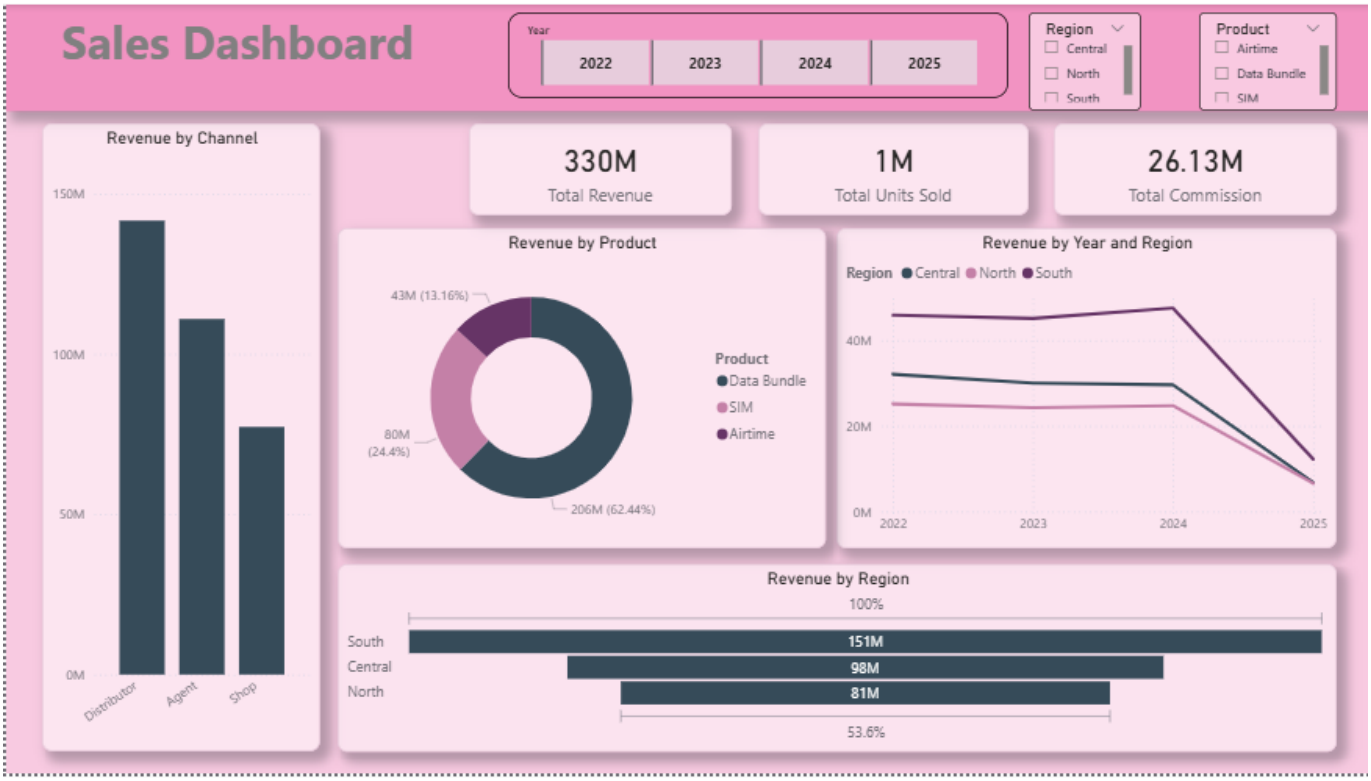
Data bundles contribute the largest share of revenue, followed by airtime, while SIM sales contribute the least. This reflects increasing demand for mobile data services.

7. Dashboard Preview

7.1 Excel interactive dashboard



7.2 Power BI dashboard



8. Key Insights

- Revenue has shown consistent growth over the years, although dropped in 2025
- The Southern region is the strongest market
- Distributors are the most effective sales channel
- Data bundles are the main driver of revenue
- Commission structures differ across channels and can impact profitability

9. Recommendations

Based on the analysis:

- Increase investment in high-performing regions such as the South
- Strengthen distributor networks to maximize sales
- Improve performance of underperforming channels such as shops
- Promote data bundle products to sustain revenue growth
- Optimize commission structures to improve efficiency